ITEP 414 – SYSTEM ADMINISTRATION AND MAINTENANCE

WEEK 2 PORTFOLIO PROJECT

ENTERPRISE INFRASTRUCTURE PLAN — NEXORA TECHNOLOGIES

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Program: Bachelor of Science in Information Technology

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PART 1 — COMPANY PROFILE

Item	Details
Company Name	Nexora Technologies
Nature of Business	Software Development & Digital Solutions — builds custom business applications, web portals, and productivity tools for small and growing enterprises across the Philippines
Vision	To be the most trusted technology partner of small and growing businesses — turning everyday challenges into simple, reliable, and future-ready digital solutions.
Mission	Deliver practical, secure, and well-crafted software; keep infrastructure straightforward but scalable; and ensure every team member has the right tools to do their best work.
Office Location	Unit 501, Prime Tech Hub, Molino Boulevard, Bacoor City, Cavite

Item	Details
Organizational Structure	Chief Executive Officer → Department Heads → Staff
Employee Distribution	IT Department: 5 · Human Resources: 4 · Finance: 5 · Sales: 6 · **Total: 20 Employees**

PART 2 — ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Quantity	Department	Purpose & Justification
H-001	Desktop PC (Core i5, 16GB RAM, 512GB SSD)	15	HR, Finance, Sales, IT Admin	Standard workstations — balanced performance and cost for daily office tasks
H-002	Laptop (Core i7, 16GB RAM, 512GB SSD)	5	IT Department	For developers & admins — portability plus power for coding, testing, and remote work
H-003	Tower Server (Xeon, 32GB RAM, 4×2TB HDD RAID 5)	1	IT Department	Central file storage, internal services, authentication — RAID 5 protects data if a drive fails
H-004	Gigabit Router	1	IT Department	Internet gateway, traffic management, VPN access
H-005	24-Port Gigabit Switch	2	IT Department	Wired backbone — stable, fast connections for all devices

Asset ID	Hardware	Quantity	Department	Purpose & Justification
H-006	Multifunction Laser Printer	1	HR + Finance	Print, scan, copy contracts, receipts, and documents
H-007	1500VA UPS	4	Server & Network Rack	Prevents data corruption and hardware damage during power interruptions
H-008	Wi-Fi 6 Wireless Access Point	3	All Departments	Wireless coverage — workstations, meeting rooms, visitors
H-009	4-Bay NAS (8TB Total)	1	IT Department	Shared storage, versioned backups, file archiving
H-010	4TB External HDD	2	IT Department	Weekly offline backups — one stored offsite
H-011	24" IPS Monitor	20	All Employees	Uniform display — comfortable for all-day work, easy replacement

Justification: 15 desktops + 5 laptops = exactly 20 employees. All mandatory equipment included with realistic quantities. RAID 5 + UPS + backup drives ensure reliability.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

Software	Version	License Type	Purpose
Windows 11 Pro	22H2	Business OEM	Primary workstation OS — domain join, BitLocker, remote management

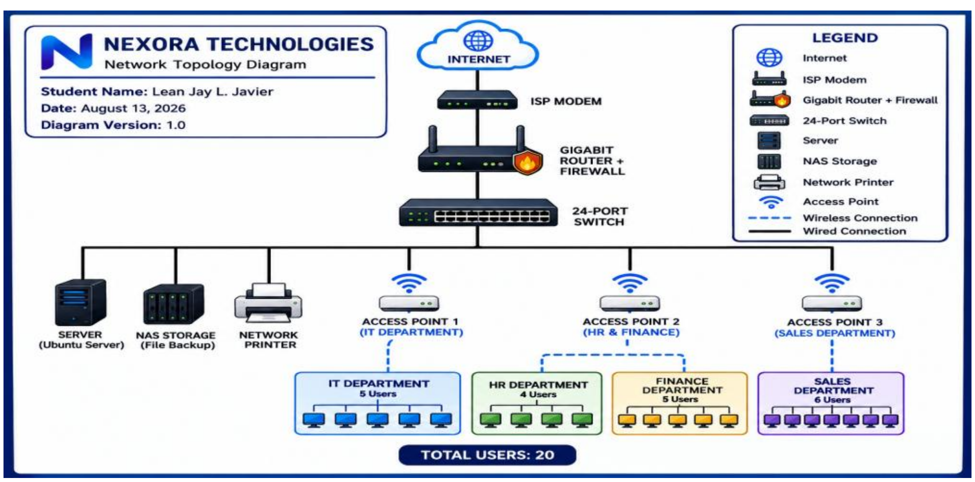
Software	**Version**	**License Type**	**Purpose**
Ubuntu Server	24.04 LTS	Free Open Source	Server OS — stable, secure, industry standard, no license cost
Microsoft Office / 365 Apps	Latest	Business Subscription	Word, Excel, Outlook, email, shared calendars — essential for all departments
Visual Studio Code	Latest	Free	Primary code editor for developers
Git	2.47.x	Free	Version control — track all code changes
GitHub Desktop	Latest	Free	Visual Git tool — simplifies team collaboration
VirtualBox	7.x	Free	Virtualization — test servers & apps without extra hardware
Google Chrome	Latest	Free	Standard browser — web apps, testing, research
Microsoft Defender	Built-in	Included	Real-time antivirus — regularly updated, no extra cost
AnyDesk	Latest	Free / Business	Remote support — assist users or manage systems from anywhere
7-Zip	24.x	Free	File compression — package deliverables, save storage space

Justification: Free/open-source tools keep startup costs low. Windows + 365 ensure business compatibility. Defender provides solid baseline protection.

PART 4 — ENTERPRISE NETWORK INVENTORY

Asset ID	Equipment	Quantity	Purpose
N-001	Fiber Internet Modem	1	ISP termination — entry point for internet connection
N-002	Gigabit Router with Firewall	1	Routing, NAT, port filtering, VPN remote access
N-003	24-Port Gigabit Switch	2	Wired network backbone — all workstations & servers connect here
N-004	Wi-Fi 6 Access Point	3	Wireless coverage — separate networks for staff and guests
N-005	24-Port Patch Panel	2	Organized cabling — easy port management & future changes
N-006	CAT6 Ethernet Cable	1 Box	Gigabit wired connections — future-proof infrastructure
N-007	RJ45 Connectors	1 Pack	Cable termination — custom-length cables as needed

PART 5 — ENTERPRISE NETWORK DIAGRAM



PART 6 — SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

- Responsibilities: Resolve desktop/laptop issues, reset passwords, install software, assist all employees, maintain support records
- Skills: Windows troubleshooting, basic networking, remote support, clear communication
- Common Tools: AnyDesk, ticketing system, USB recovery media, diagnostic utilities
- Certifications: CompTIA A+, Microsoft Certified: Support Technician Associate

Network Administrator

- Responsibilities: Manage router, switch, firewall, and Wi-Fi; maintain internet & internal connections; monitor performance; troubleshoot connectivity
- Skills: IP addressing, subnetting, DHCP/DNS, firewall rules, cable termination
- Common Tools: Ping/traceroute, cable tester, Wi-Fi analyzer, router admin interface
- Certifications: CompTIA Network+, CCNA

Linux System Administrator

- Responsibilities: Install, configure, and maintain Ubuntu Server; manage user accounts, permissions, services, backups; apply security updates; monitor system health
- Skills: Command line, file permissions, shell scripting, RAID, UFW firewall
- Common Tools: SSH, nano/vim, rsync, systemd
- Certifications: CompTIA Linux+, LPIC-1

Cloud Administrator

- Responsibilities: Manage cloud services, secure remote access, offsite backups, identity policies, future cloud integration
- Skills: Cloud basics, VPN, Multi-Factor Authentication, cost monitoring
- Common Tools: Azure/AWS console, cloud backup tools, MFA management platforms
- Certifications: CompTIA Cloud Essentials+, Microsoft Azure Fundamentals

How These Professionals Work Together

The Helpdesk Technician serves as the first point of contact for all staff — resolving daily issues and escalating complex or recurring problems to specialists. The Network Administrator ensures that internet, internal connections, and remote access remain reliable so everyone can reach servers and cloud resources. The Linux System Administrator maintains the actual server environment, safeguards stored data, and applies consistent security updates. The Cloud Administrator designs secure remote access solutions and manages offsite backups so the team can work safely from anywhere. All four roles share documentation, coordinate maintenance schedules, and align on security policies — together they keep the entire infrastructure consistent, protected, and easy to support.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Choose a business fiber plan (PLDT or Globe Business, minimum 300Mbps) with static IP. Static IP enables secure remote access for developers and admins. Business-grade plans offer better uptime guarantees and faster support compared to residential plans.

Server Specifications

Tower server with Xeon processor, 32GB RAM, and RAID 5 (4×2TB HDD). RAID 5 provides fault tolerance — if one drive fails, data remains accessible. Use Ubuntu Server 24.04 LTS — free, secure, well-supported, and requires fewer resources than paid server operating systems.

Backup Strategy

Follow the 3-2-1 Backup Rule: maintain 3 copies of data, on 2 different media types, with 1 copy stored offsite. Schedule automatic daily backups to the NAS, and perform a full weekly backup to an external drive — one kept in the office, one stored at a secure alternate location.

Security Recommendations

- Antivirus: Enable Microsoft Defender real-time protection, cloud-delivered protection, and scheduled weekly scans. Built-in, regularly updated, and sufficient when paired with good user habits.
- Password Policy: Minimum 10 characters (mix letters, numbers, symbols). Change every 90 days. No password sharing. Enable Multi-Factor Authentication (MFA) on email, cloud accounts, and remote access.

- Expansion Plan: Start with wired connections for desktops — faster and more secure. As the company grows, add Wi-Fi access points, increase storage, and gradually migrate services to the cloud. Standardize hardware models for easier replacements and upgrades.

PART 8 — PERSONAL REFLECTION

Working on this infrastructure plan taught me that system administration is not only about fixing computers or using commands. I realized that planning and understanding what a company really needs are also very important. Before doing this project, I thought setting up an office was mostly about buying computers, network devices, and connecting them. As I worked on the project, I learned that every hardware, software, and network setup should have a purpose and should help the company perform its work better.

The most challenging part for me was deciding what equipment and specifications would be appropriate for a startup company. At first, I wanted to choose high-end equipment because I thought it would be better. However, I realized that a startup also needs to consider its budget. I had to think about whether a certain device was really necessary, if there was a simpler option, and if it could still handle the company's needs. This helped me understand that choosing IT equipment is not always about getting the most expensive or powerful option. It is about finding the right balance between cost, performance, and future needs.

I also learned why planning is important before purchasing and setting up equipment. A good infrastructure plan can help prevent unnecessary expenses and problems in the future. If the wrong equipment is purchased or the network is poorly designed, it can be difficult and expensive to fix later. Proper documentation is also important because it makes it easier for system administrators to understand, maintain, and troubleshoot the infrastructure.

Overall, this project helped me develop a better understanding of what a System Administrator does. It taught me to think carefully before making technical decisions and to document information properly. I also learned that technical decisions should be explained in a way that other people, including non-technical managers, can understand. Most importantly, I learned that a good IT infrastructure does not have to be complicated or expensive. It should be reliable, secure, easy to maintain, and most importantly, suitable for the actual needs of the company.